Zahin

November 19, 2025

[ ]:
```python
import pandas as pd

df = pd.read_csv("data/brca_data_w_subtypes.csv")
print(df.shape)
print(df.columns[:10])
df.head()
```

```
(705, 1941)
Index(['rs_CLEC3A', 'rs_CPB1', 'rs_SCGB2A2', 'rs_SCGB1D2', 'rs_TFF1',
       'rs_MUCL1', 'rs_GSTM1', 'rs_PIP', 'rs_ADIPOQ', 'rs_ADH1B'],
      dtype='object')
```

[ ]:

	rs_CLEC3A	rs_CPB1	rs_SCGB2A2	rs_SCGB1D2	rs_TFF1	rs_MUCL1	\
0	0.892818	6.580103	14.123672	10.606501	13.189237	6.649466	
1	0.000000	3.691311	17.116090	15.517231	9.867616	9.691667	
2	3.748150	4.375255	9.658123	5.326983	12.109539	11.644307	
3	0.000000	18.235519	18.535480	14.533584	14.078992	8.913760	
4	0.000000	4.583724	15.711865	12.804521	8.881669	8.430028	

	rs_GSTM1	rs_PIP	rs_ADIPOQ	rs_ADH1B	…	pp_p62.LCK.ligand	\
0	10.520335	10.338490	10.248379	10.229970	…	-0.691766	
1	8.179522	7.911723	1.289598	1.818891	…	0.279067	
2	10.517330	5.114925	11.975349	11.911437	…	0.219910	
3	10.557465	13.304434	8.205059	9.211476	…	-0.266554	
4	12.964607	6.806517	4.294341	5.385714	…	-0.441542	

	pp_p70S6K	pp_p70S6K.pT389	pp_p90RSK	pp_p90RSK.pT359.S363	vital.status	\
0	-0.337863	-0.178503	0.011638	-0.207257	0	
1	0.292925	-0.155242	-0.089365	0.267530	0	
2	0.308110	-0.190794	-0.222150	-0.198518	0	
3	-0.079871	-0.463237	0.522998	-0.046902	0	
4	-0.152317	0.511386	-0.096482	0.037473	0	

	PR.Status	ER.Status	HER2.Final.Status	histological.type
0	Positive	Positive	Negative	infiltrating ductal carcinoma
1	Positive	Negative	Negative	infiltrating ductal carcinoma
2	Positive	Positive	Negative	infiltrating ductal carcinoma
3	Positive	Positive	Negative	infiltrating ductal carcinoma

```
4   Positive   Positive           Negative  infiltrating ductal carcinoma

[5 rows x 1941 columns]
```

[ ]:
```python
non_numeric = df.select_dtypes(include='object').columns
print("Non-numeric columns:")
print(non_numeric)
```

```
Non-numeric columns:
Index(['PR.Status', 'ER.Status', 'HER2.Final.Status', 'histological.type'],
dtype='object')
```

[ ]:
```python
print(df['ER.Status'].value_counts())
```

```
ER.Status
Positive                       414
Negative                       135
Not Performed                   27
Performed but Not Available      5
Indeterminate                    2
Name: count, dtype: int64
```

[ ]:
```python
import numpy as np

# Filtering to only Positive/Negative ER status
df_clean = df[df['ER.Status'].isin(['Positive', 'Negative'])].copy()

# Drop other metadata columns
df_clean = df_clean.drop(['PR.Status', 'HER2.Final.Status', 'histological.
 ↪type'], axis=1)

# Split into features and labels
X = df_clean.drop('ER.Status', axis=1)
y = np.where(df_clean['ER.Status'] == 'Positive', 1, 0)

print("X shape:", X.shape)
print("y shape:", y.shape)
print("Positive samples:", y.sum())
```

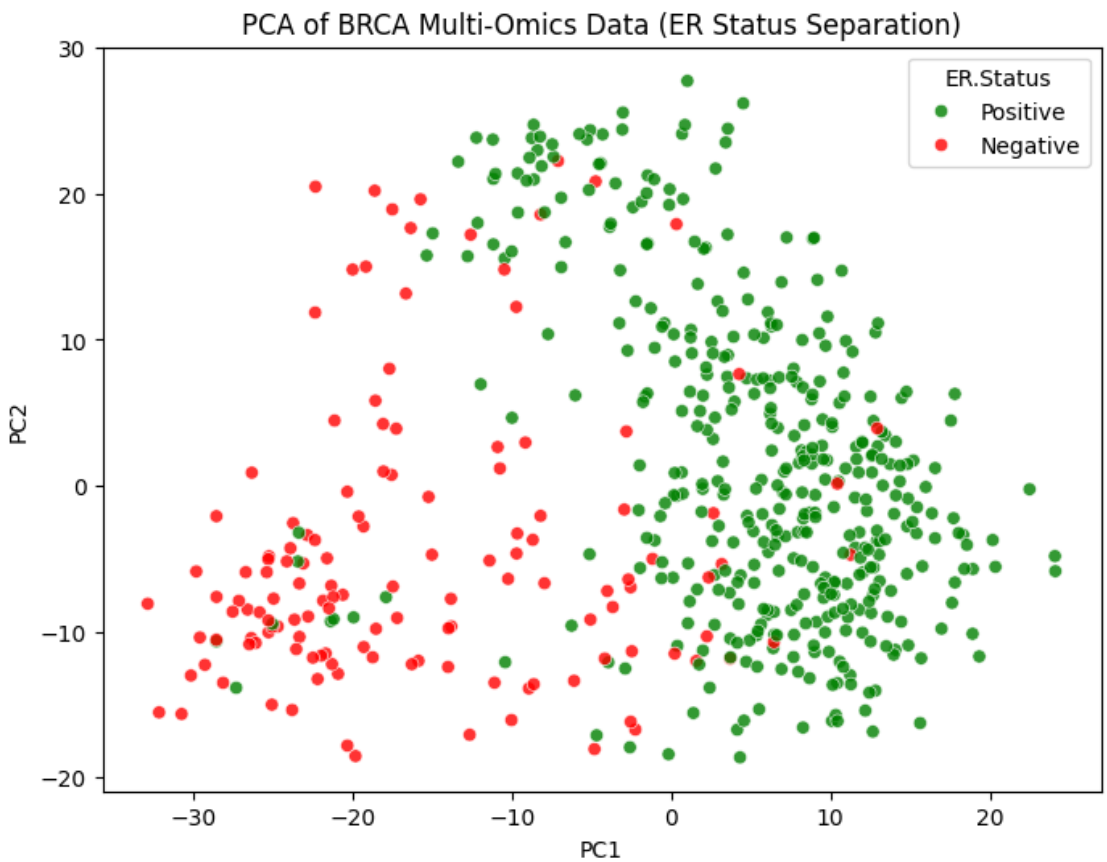
```
X shape: (549, 1937)
y shape: (549,)
Positive samples: 414
```

[ ]:
```python
from sklearn.preprocessing import StandardScaler

# Replacing any missing values (NaNs) with the feature median
X_filled = X.fillna(X.median())

# Scale (z-score normalization)
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_filled)

print("Scaled data shape:", X_scaled.shape)
print("Mean of first 5 features:", X_scaled[:, :5].mean(axis=0))
print("Std of first 5 features:", X_scaled[:, :5].std(axis=0))
```

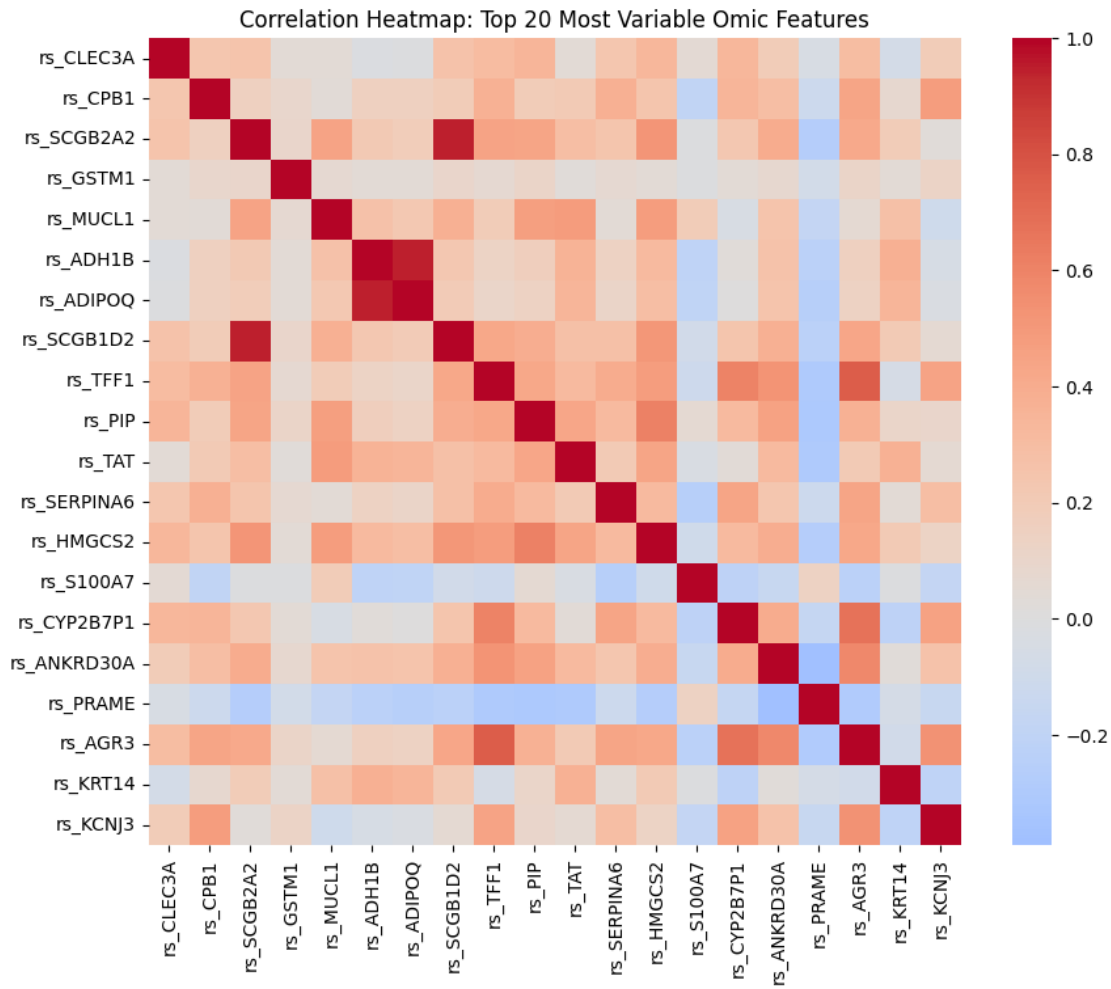
```
Scaled data shape: (549, 1937)
Mean of first 5 features: [-3.55918492e-17 -4.85343399e-17 -2.07079850e-16
-1.55309888e-16
  2.36200454e-16]
Std of first 5 features: [1. 1. 1. 1. 1.]
```

[ ]:
```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.decomposition import PCA

# Run PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# dataframe for plotting
pca_df = pd.DataFrame({
    'PC1': X_pca[:, 0],
    'PC2': X_pca[:, 1],
    'ER.Status': ['Positive' if val == 1 else 'Negative' for val in y]
})

# Scatter plot
plt.figure(figsize=(8,6))
sns.scatterplot(data=pca_df, x='PC1', y='PC2', hue='ER.Status',
                palette={'Positive':'green','Negative':'red'}, alpha=0.8)
plt.title('PCA of BRCA Multi-Omics Data (ER Status Separation)')
plt.show()

# Explained variance
print("Explained variance ratio:", pca.explained_variance_ratio_)
```




```
Explained variance ratio: [0.08278482 0.06188028]
```

[ ]:
```python
# Compute variance per feature
variances = X.var().sort_values(ascending=False)

# Select top 20 most variable features
top_features = variances.head(20).index

# Compute correlation matrix
corr = X[top_features].corr()

# Plot heatmap
plt.figure(figsize=(10,8))
sns.heatmap(corr, cmap='coolwarm', center=0)
plt.title("Correlation Heatmap: Top 20 Most Variable Omic Features")
plt.show()
```

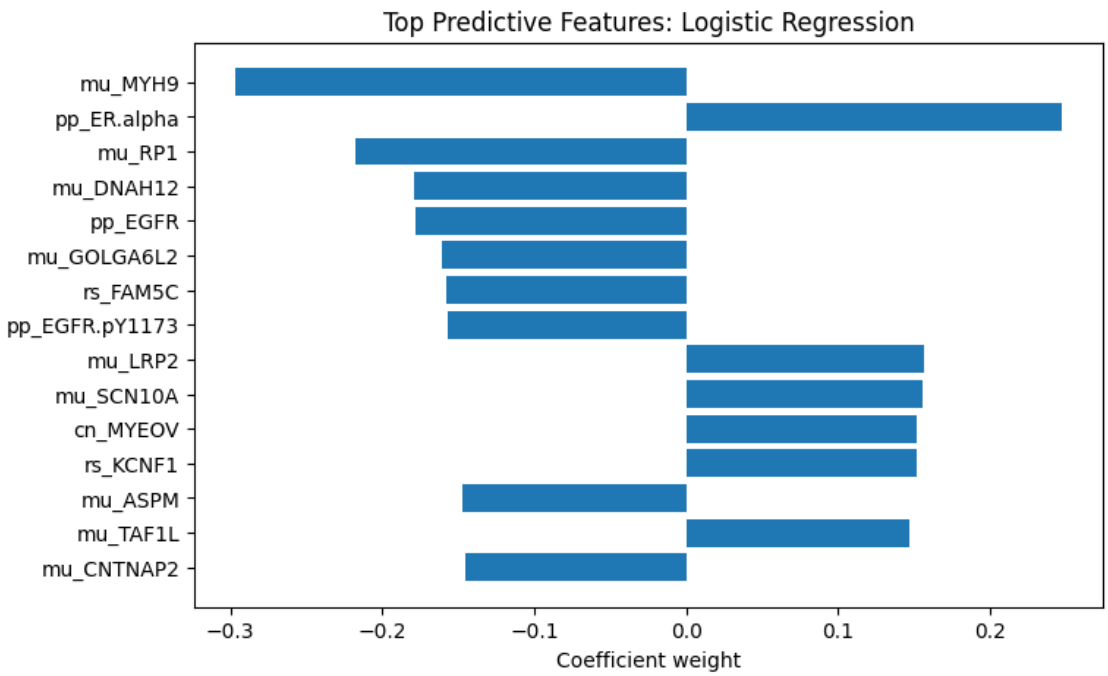


[ ]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report,␣
↪confusion_matrix

# Split data (80 / 20)
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, stratify=y, random_state=42)

# Train baseline model
lr = LogisticRegression(max_iter=1000, random_state=42)
lr.fit(X_train, y_train)

# Evaluate
y_pred = lr.predict(X_test)
acc = accuracy_score(y_test, y_pred)
```

```python
print("Baseline Logistic Regression Accuracy:", round(acc, 3))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Baseline Logistic Regression Accuracy: 0.909

Classification Report:
               precision    recall  f1-score   support

           0       0.81      0.81      0.81        27
           1       0.94      0.94      0.94        83

    accuracy                           0.91       110
   macro avg       0.88      0.88      0.88       110
weighted avg       0.91      0.91      0.91       110
```

[ ]:
```python
import numpy as np
coef = lr.coef_[0]
top_idx = np.argsort(np.abs(coef))[-15:][::-1]
top_features = X.columns[top_idx]
top_values = coef[top_idx]

import matplotlib.pyplot as plt
plt.figure(figsize=(8,5))
plt.barh(top_features, top_values)
plt.gca().invert_yaxis()
plt.title("Top Predictive Features: Logistic Regression")
plt.xlabel("Coefficient weight")
plt.show()
```


Top Predictive Features: Logistic Regression
mu_MYH9
pp_ER.alpha
mu_RP1
mu_DNAH12
pp_EGFR
mu_GOLGA6L2
rs_FAM5C
pp_EGFR.pY1173
mu_LRP2
mu_SCN10A
cn_MYEOV
rs_KCNF1
mu_ASPM
mu_TAF1L
mu_CNTNAP2
−0.3
−0.2
−0.1
0.0
0.1
0.2
Coefficient weight

```
[ ]:
```